

SMTP (Simple Mail Transfer Protocol) – Simple Steps

1. What is a Mail Server?

A Mail Server is a computer (server) that sends, receives, stores, and delivers emails

Package Name : postfix

Configuration file : /etc/postfix/main.cf

Port number : 25

Service name : postfix.service

Install package:

yum install postfix

```
[root@server ~]# yum install postfix -y
Last metadata expiration check: 4:53:25 ago on Sun 05 Jul 2026 11:46:41 AM IST.
Dependencies resolved.
=====
Package                Architecture          Version               Repository
=====
Installing:
postfix                x86_64                2:3.5.25-2.el9       appstream

Transaction Summary
=====
Install 1 Package
```

Start and enable Postfix so it runs after a reboot:

systemctl enable --now postfix.service

```
[root@server ~]# systemctl enable --now postfix.service
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
```

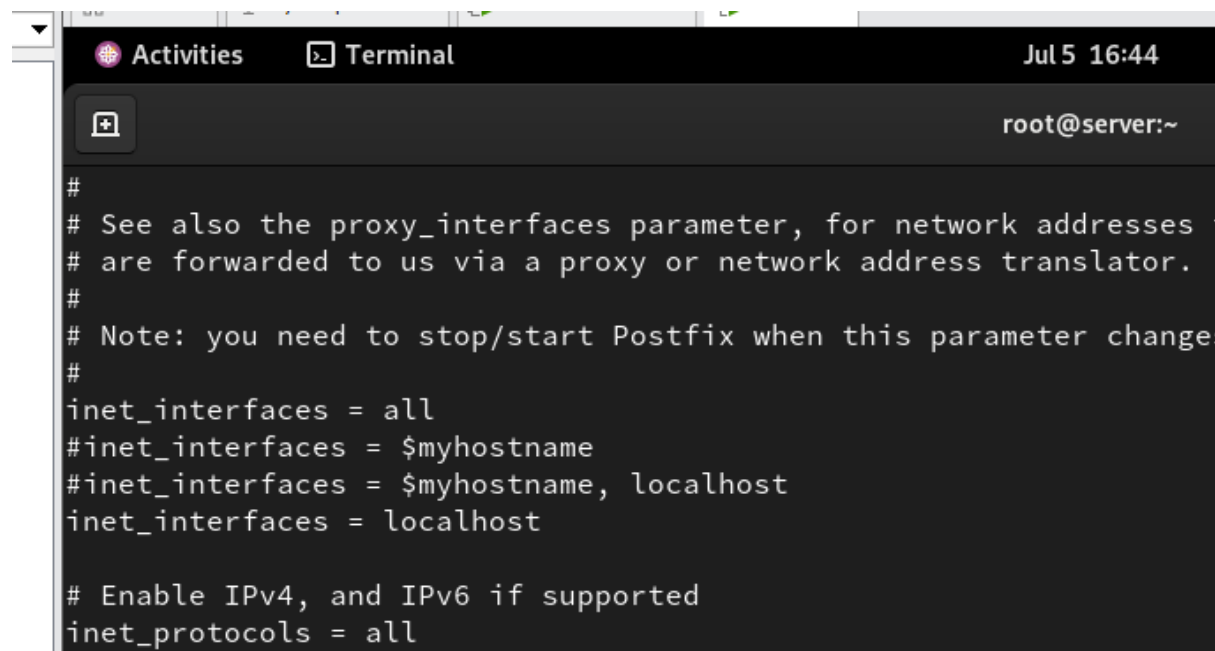
```
[root@server ~]# systemctl status postfix.service
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-07-05 16:41:32 IST; 33s ago
     Process: 13322 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix (code=0)
     Process: 13323 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=0)
     Process: 13327 ExecStartPre=/usr/libexec/postfix/chroot-update (code=0)
```

Edit configuration file:

#vi /etc/postfix/main.cf

```
[root@server ~]# vi /etc/postfix/main.cf
```

REMOVE THE # SYMBOL

A screenshot of a Linux terminal window. The window title bar shows 'Activities', 'Terminal', and the date 'Jul 5 16:44'. The prompt is 'root@server:~'. The terminal displays the following configuration for 'inet_interfaces':

```
#  
# See also the proxy_interfaces parameter, for network addresses  
# are forwarded to us via a proxy or network address translator.  
#  
# Note: you need to stop/start Postfix when this parameter change  
#  
inet_interfaces = all  
#inet_interfaces = $myhostname  
#inet_interfaces = $myhostname, localhost  
inet_interfaces = localhost  
  
# Enable IPv4, and IPv6 if supported  
inet_protocols = all
```

Restart the service

systemctl restart postfix.service

```
[root@server ~]# systemctl restart postfix.service
```

Create a user

useradd

```
[root@server ~]# cd /home  
[root@server home]# ls  
hari niva raj tamil  
[root@server home]# passwd niva  
Changing password for user niva.  
New password:  
BAD PASSWORD: The password is a palindrome  
Retype new password:  
passwd: all authentication tokens updated successfully.
```

Sending mail to a user:

```
# mail -s "Test Email" mail@localhost
```

```
[root@server home]# mail -s "server to user" niva
To: niva
Subject: server to user

hi am raj
i am 21
we are all
^D
-----
(Preliminary) Envelope contains:
To: niva
Subject: server to user
Send this message [yes/no, empty: recompose]? y
sendmail: warning: /etc/postfix/main.cf, line 135: overriding earlier entry: inet_interfaces=all
postdrop: warning: /etc/postfix/main.cf, line 135: overriding earlier entry: inet_interfaces=all
[root@server home]#
```

To check logs of mail:

```
# tail /var/log/maillog
```

```
[root@server home]# tail -f /var/log/maillog
Jul  5 16:48:48 client postfix/sendmail[13630]: warning: /etc/postfix/main.cf, line 135: overriding earlier entry:
inet_interfaces=all
Jul  5 16:48:48 client postfix/postdrop[13632]: warning: /etc/postfix/main.cf, line 135: overriding earlier entry:
inet_interfaces=all
Jul  5 16:48:48 client postfix/cleanup[13633]: warning: /etc/postfix/main.cf, line 135: overriding earlier entry:
inet_interfaces=all
Jul  5 16:48:48 client postfix/pickup[13574]: 73E4432BCF2D: uid=0 from=<root>
Jul  5 16:48:48 client postfix/trivial-rewrite[13634]: warning: /etc/postfix/main.cf, line 135: overriding earlier
entry: inet_interfaces=all
Jul  5 16:48:48 client postfix/cleanup[13633]: 73E4432BCF2D: message-id=<20260705111848.73E4432BCF2D@client.locald
omain>
Jul  5 16:48:48 client postfix/qmgr[13575]: 73E4432BCF2D: from=<root@client.localdomain>, size=339, nrcpt=1 (queue
active)
Jul  5 16:48:48 client postfix/local[13635]: warning: /etc/postfix/main.cf, line 135: overriding earlier entry: in
et_interfaces=all
Jul  5 16:48:48 client postfix/local[13635]: 73E4432BCF2D: to=<niva@client.localdomain>, orig_to=<niva>, relay=loc
al, delay=0.18, delays=0.11/0.07/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Jul  5 16:48:48 client postfix/qmgr[13575]: 73E4432BCF2D: removed
```

Check mail is send to another user:

```
# su - mailuser
```

```
# mail
```

```
[root@server home]# su - niva
[niva@client ~]$
```

```
[niva@client ~]$ mail
s-nail version v14.9.22.  Type `?' for help
/var/spool/mail/niva: 2 messages 1 new
  1 root                2026-07-05 16:48    15/494    "server to user
>N  2 root                2026-07-05 16:52    17/514    "server to user
```

Command	Purpose	Example
p	Print current email	p
h or headers	Show email list	h
n	Next email	n
d	Delete current email	d
u	Undelete deleted email	u
s filename	Save email to a file	s mail.txt
mail user	Send a new email	mail niva
r	Reply to current email	r
q	Quit and save changes	q
x	Exit without saving changes	x
! command	Run a Linux command	!pwd

#Mail

```
[root@client ~]# mail
s-nail version v14.9.22.  Type '?' for help
/var/spool/mail/root: 2 messages
▶ 1 niva@client.localdom 2026-07-05 16:59 25/780 "Re: server to user
  2 niva@client.localdom 2026-07-05 17:02 17/635 "*** SECURITY information for client
&
```

#1 list the first mail

```
& 1
[-- Message 1 -- 25 lines, 780 bytes --]:
Date: Sun, 05 Jul 2026 16:59:22 +0530
To: root <root@client.localdomain>
Subject: Re: server to user
Cc: niva@client.localdomain
Message-Id: <20260705112922.614F132BCF17@client.localdomain>
From: niva@client.localdomain

root <root@client.localdomain> wrote:

> hi am raj
> i am 21
> we are all
thanks you al
wish all the best
```

#h

```
& h
▶ 1 niva@client.localdom 2026-07-05 16:59 25/780 "Re: server to user
  2 niva@client.localdom 2026-07-05 17:02 17/635 "*** SECURITY information
```

Create one user and add multiple users in config file:

vi /etc/aliases

```
password.  
[root@client ~]# vi /etc/aliases  
  
#  
# Aliases in this file will NOT be expanded in the header from  
# Mail, but WILL be visible over networks or from /bin/mail.  
#  
# >>>>>>>>> The program "newaliases" must be run after  
# >> NOTE >> this file is updated for any changes to  
# >>>>>>>>> show through to sendmail.  
#  
# Basic system aliases -- these MUST be present.  
mailer-daemon: postmaster  
postmaster: root  
group:niva,raj,hari  
  
# General redirections for pseudo accounts.  
bin: root
```

Save the file and update aliases:

newaliases

```
[root@client home]# newaliases  
newaliases: warning: /etc/postfix/main.cf, line 135: overriding earlier entry: inet_interfaces=all  
postalias: warning: /etc/postfix/main.cf, line 135: overriding earlier entry: inet_interfaces=all
```

Send mail to a group:

mail -s "subject" groupname

```
[root@client home]# mail -s "info" group  
To: group  
Subject: info  
  
hi am  
raj  
we all foir  
^D  
-----  
(Preliminary) Envelope contains:  
To: group  
Subject: info  
Send this message [yes/no, empty: recompose]? y  
sendmail: warning: /etc/postfix/main.cf, line 135: overriding earlier entry: inet_interfaces=all  
postdrop: warning: /etc/postfix/main.cf, line 135: overriding earlier entry: inet_interfaces=all
```

Check mail is send to user1

```
[root@client home]# su - raj
[raj@client ~]$ mail
s-nail version v14.9.22.  Type '?' for help
/var/spool/mail/raj: 1 message 1 new
•N 1 root                2026-07-05 17:24    17/501    "info
& 1
[-- Message 1 -- 17 lines, 501 bytes --]:
Date: Sun, 05 Jul 2026 17:24:32 +0530
To: group@client.localdomain
Subject: info
Message-Id: <20260705115432.17EB232BCF17@client.localdomain>
From: root <root@client.localdomain>

hi am
raj
we all foir

& q
```

User2

```
[raj@client ~]$ su - niva
Password:
[niva@client ~]$ mail
s-nail version v14.9.22.  Type '?' for help
/var/spool/mail/niva: 4 messages 1 new 2 unread
  1 root                2026-07-05 16:48    15/494    "server to user
A 2 root                2026-07-05 16:52    19/537    "server to user
U 3 niva@client.localdom 2026-07-05 16:59    25/779    "Re: server to use
•N 4 root                2026-07-05 17:24    17/501    "info
& 4
[-- Message 4 -- 17 lines, 501 bytes --]:
Date: Sun, 05 Jul 2026 17:24:32 +0530
To: group@client.localdomain
Subject: info
Message-Id: <20260705115432.17EB232BCF17@client.localdomain>
From: root <root@client.localdomain>

hi am
raj
we all foir

& q
```

User3

```
[niva@client ~]$ su - hari
Password:
[hari@client ~]$ mail
s-nail version v14.9.22.  Type '?' for help
/var/spool/mail/hari: 1 message 1 new
•N  1 root                2026-07-05 17:24    17/501    "info
& 1
[-- Message  1 -- 17 lines, 501 bytes --]:
Date: Sun, 05 Jul 2026 17:24:32 +0530
To: group@client.localdomain
Subject: info
Message-Id: <20260705115432.17EB232BCF17@client.localdomain>
From: root <root@client.localdomain>

hi am
raj
we all foir
```